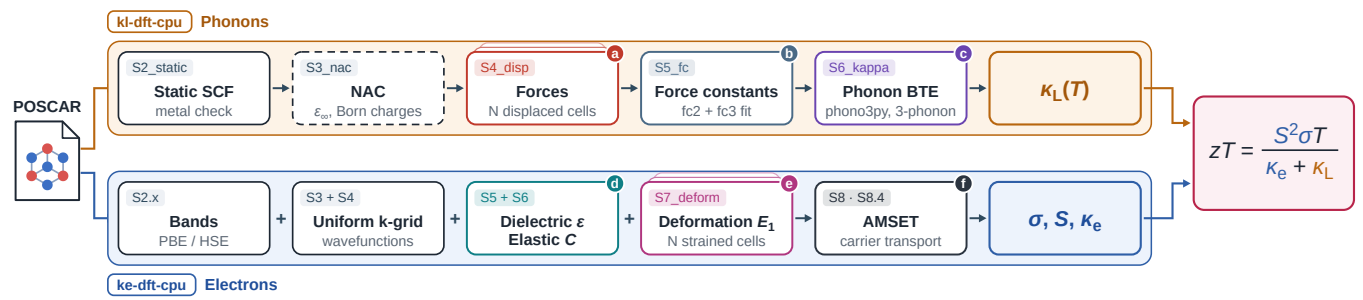


a Random displacements

S4_disp · METHOD = alm (default)



All atoms rattled, RMS 0.03 Å

***N* frames** ≈ n_{free} / $3N_{sc}$ × 3

n_{free} from ALM; $10 \le N \le 500$

b Force-constant fit GPU

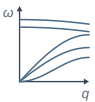
S5_fc · FIT_ENGINE = pheasy (ALASSO)

fc2 + fc3 fitted on the GPU

GPU matvec = CPU to ~10^{-16}

Gate: no modes below −0.1 THz

2D: ZA ∝ q^p, $1.7 < p < 2.3$



c Phonon BTE

S6_kappa · phono3py, BTE_METHOD = rta

3-phonon BTE, RTA (or LBTE)

q-mesh converged automatically

Cutoff = largest on the κ_L plateau

Isotopes on; **NAC** off for 2D



d Scattering inputs

S5_dielect · S6_elastic

Dielectric ε

3D DFPT ε_0, ε_∞

2D in-plane slab ε, 2D screening

Elastic C

3D C_{ij} in GPa, Voigt order

2D in-plane 2 × 2, $C_{2D} = c \cdot C_{slab}$

e Deformation potential E_1

S7_deform → S7.1_read · DEFORM_REF = auto

3D reference: core level

2D reference: vacuum level (LOCPOT)



ions relaxed: all strains (3D), in-plane strains (2D)

f Carrier transport

S8_kappa · S8.4_amset2d

ADP POP IMP PIE ODP *scattering*

3D **AMSET**, 3D kernels

2D **AMSET**, 2D kernels

3D 2D **BoltzTraP2** CRTA × DPT check

S8.1_bt2 + S8.2_dpt, on by default